

SYSTEM MODULES

AI Modules Overview

AgriXAI — Cognitive Reasoning Layer

Dual-head **EfficientNet-B3** for hierarchical disease, pest, and nutrient deficiency classification. Accuracy: **92.56%**. Enriched with Grad-CAM visual explanations and BCR severity grading.

AgriChat Bilingual Hybrid RAG System

FAISS semantic search over **28,613** agricultural paragraphs fused with Knowledge Graph traversal. Score: **0.803** vs. 0.697 baseline (+15.21%).

AI AgriVision Fertilization Recommendation System

Mistral-7B fine-tuned via **QLoRA** delivering NPK & pH recommendations calibrated against Egypt's Ministry of Agriculture standards.

Egypt FAO Crop Yield Prediction System

Random Forest + Gradient Boosting ensemble for **68 Egyptian crops** across 6 climate scenarios. $R^2 = 0.943$.

EWS-PA v2.3: Vegetation Stress Early Warning System

Satellite NDVI monitoring of **31 Egyptian nature reserves** for proactive vegetation stress detection. $ROC-AUC = 0.817$.

MOTIVATION & CONTEXT

The Information Gap & Economic Impact

The Challenge

Egyptian farmers face critical information barriers that lead to the loss of **20%–50%** of crop yields annually due to delayed disease diagnosis and lack of localized advisory tools.

8–12B EGP

Annual preventable economic losses in Egyptian agriculture

The Goal

Deliver a comprehensive AI-powered advisory platform operating **100% Offline-First** — no internet required — ensuring data privacy, instant response, and maximum accessibility across rural Egypt.



GRADUATION PROJECT 2025–2026

AI AgriVision

An integrated AI system for smart agricultural advisory, crop disease diagnosis, and yield prediction — fully offline.

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METHODOLOGY & ARCHITECTURE

How the Systems Were Built

Fine-Tuning Architecture — QLoRA

QLoRA reduces memory consumption by **4×** via **4-bit NF4** weight quantization, training only low-rank matrices **A & B** while keeping the base model frozen — enabling Mistral-7B to run locally without cloud GPUs.

AgriChat Knowledge Graph

4,243

Nodes

4,233

Edges

777

Communities

Partitioned via the **Louvain algorithm** for efficient multi-hop semantic traversal.

AgriXAI Diagnosis Pipeline

- 1 **Image Input** — leaf photo capture
- 2 **EfficientNet-B3** — dual-head classification
- 3 **Grad-CAM** — infection zone mapping
- 4 **BCR Grading** — economic severity scoring
- 5 **AgriChat RAG** — treatment recommendation

AgriXAI Module 1 — Reasoning Graph

17

Categories

90

Nodes

120

Directed Edges

Directed edges map pathogenic causes to economic treatment recommendations.

AGRICULTURAL IMPACT & PROJECTIONS

Impact & Future Projections (2025–2035)

Crop Productivity Increase

+20%

Expected

+35%

Optimistic

Wheat, rice, tomatoes & corn — under climate scenario modeling.

Agricultural Waste Reduction

-32.5%

Preventable crop losses reduced via early BCR-based intervention.

Fertilizer & Pesticide Cost Savings

-20%

Via precision NPK & pH QLoRA recommendations.

Food Security Contribution

Early-warning EWS-PA covers **31 protected areas** preventing ecosystem degradation, directly safeguarding agricultural buffer zones across the Nile Delta and Upper Egypt.

MENA AI Adoption in Agriculture

18%

2025



62%

2035

AI AgriVision is positioned at the heart of this market opportunity.

COMPETITIVE EDGE

Why AI AgriVision?



Explainable AI — Farmers Trust the Why

Grad-CAM heatmaps and BCR economic grading ensure farmers don't just get an answer — they understand it and can justify it to agricultural engineers.



Egyptian-Specific by Design

Trained on Egypt FAO data, calibrated to Ministry of Agriculture standards, covering 15+ local crops and 31 Egyptian nature reserves — not a generic global tool.



6 Modules — One Unified Platform

From disease diagnosis to yield prediction, fertilization advisory to ecological early warning — all integrated via a **FastAPI** microservice layer and a single **React + Vite** interface.



Privacy-First Architecture

All farmer data, crop images, and advisory history remain on-device. No cloud upload, no third-party data sharing — fully compliant with agricultural data sovereignty principles.

Bilingual — Arabic & English

Full Arabic RTL support across all modules makes AI AgriVision accessible to every Egyptian farmer, regardless of language background or technical literacy.